

## Why Non-Linear Activation Functions are Essential

A neural network **must** use **non-linear activation functions** (like ReLU or sigmoid) in its hidden layers.

If a neural network only uses **linear activation functions** ( $g(z) = z$ ) in all its layers, the entire network becomes computationally equivalent to a simple **linear regression** model. This completely defeats the purpose of having multiple "deep" layers, as the network will be unable to learn complex, non-linear patterns.

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## Mathematical Proof (The "Collapse") 🐛

Let's examine a simple 2-layer network where every activation function is linear ( $g(z) = z$ ).

### 1. Layer 1 (Hidden Layer):

- The computation is:  $a_1 = g(w_1x + b_1)$
- With a linear activation, this becomes:  $a_1 = w_1x + b_1$

### 2. Layer 2 (Output Layer):

- The computation is:  $a_2 = g(w_2a_1 + b_2)$
- With a linear activation, this becomes:  $a_2 = w_2a_1 + b_2$

### 3. Substitute $a_1$ into $a_2$ :

- $a_2 = w_2(w_1x + b_1) + b_2$
- $a_2 = (w_2w_1)x + (w_2b_1 + b_2)$

### 4. Result:

- We can define a new, single  $W = w_2w_1$  and a new, single  $B = w_2b_1 + b_2$ .
- The final output is:  $a_2 = Wx + B$

This final equation is just the formula for simple linear regression. No matter how many "deep" layers you add, if they are all linear, they "collapse" into a single linear function.

**Key Principle:** A linear function of another linear function is, itself, just a linear function.

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## What About Mixed Functions?

This "collapse" still happens even if the output layer is non-linear (like sigmoid).

- If all **hidden layers are linear** ( $g(z) = z$ ) but the **output layer is sigmoid**:
  - The network collapses and becomes equivalent to a simple **logistic regression** model.
  - You gain no advantage from the "deep" architecture.
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## Conclusion & Rule of Thumb 📜

- **Hidden Layers:** You **must** use a non-linear activation function.
  - **Recommendation:** The default and most common choice is **ReLU**.
- **Output Layer:** You *can* use a linear activation function, but **only** if you are solving a regression problem (as discussed in the previous video).